

COMPUTER NETWORKS

Course Code: **23CS301** • Semester: **3-1** • Department of Computer Science & Engineering

Course Scope & Objectives: OSI & TCP/IP reference architectures, packet switching, error control, routing protocols, transport reliability, congestion control, and network application protocols.

Unit 1: Network Models & Physical Layer

Key Syllabus Topics: OSI 7-Layer Model vs TCP/IP Protocol Suite, Transmission Media (Twisted Pair, Fiber Optics, Wireless), Data Encoding, Multiplexing (FDM, TDM, WDM), Switching (Circuit vs Packet Switching), Latency, Bandwidth-Delay Product.

Theoretical Framework & Concepts:

Layered architectures modularize networking responsibilities from physical bit transmission up to end-user application semantics. Packet switching routes discrete chunks independently across shared links.

Implementation / Key Formula Reference:

```
// OSI Layers: Physical, Data Link, Network, Transport, Session, Presentation, Application
```

Frequently Asked University Exam Questions:

• **Q1: Compare OSI Model and TCP/IP Model.**

Solution: A: OSI is a 7-layer theoretical reference framework with strict layer boundaries; TCP/IP is a 4-layer practical protocol architecture that powers the modern Internet.

Unit 2: Data Link Layer & Medium Access Control (MAC)

Key Syllabus Topics: Framing, Error Detection (Parity, Checksum, CRC), Error Correction (Hamming Codes), Flow Control (Stop-and-Wait, Sliding Window Go-Back-N, Selective Repeat), Random Access Protocols (ALOHA, CSMA/CD, CSMA/CA), Ethernet & IEEE 802.11 WiFi.

Theoretical Framework & Concepts:

Data Link Layer manages node-to-node frame transfer over a single link. CRC uses polynomial binary division for reliable burst error detection. CSMA/CD senses carrier before transmitting and aborts on collision.

Implementation / Key Formula Reference:

```
// CRC Generation: Divide data polynomial  $D(x) \cdot x^r$  by generator  $G(x)$ 
```

Frequently Asked University Exam Questions:

• **Q1: Explain CSMA/CD mechanism and how collisions are handled in Ethernet.**

Solution: A: A station listens before transmitting. If collision occurs while sending, it broadcasts a jam signal, stops, and executes Binary Exponential Backoff before retransmitting.

Unit 3: Network Layer & Routing Protocols

Key Syllabus Topics: IPv4 vs IPv6 Addressing, Subnetting & CIDR, Address Resolution Protocol (ARP), Dynamic Host Configuration (DHCP), ICMP, Routing Algorithms (Distance Vector - Bellman Ford, Link State - Dijkstra), Intra-domain (OSPF, RIP) vs Inter-domain Routing (BGP), NAT.

Theoretical Framework & Concepts:

Network layer provides end-to-end host addressing and packet routing across heterogeneous networks. CIDR eliminates rigid address classes using variable length prefix masks (/24, /16).

Implementation / Key Formula Reference:

```
// Subnet mask calculation: /26 -> 255.255.255.192 (64 addresses per subnet)
```

Frequently Asked University Exam Questions:

• **Q1: Explain Count-to-Infinity problem in Distance Vector Routing and its solution.**

Solution: A: It occurs when a link breaks and nodes slowly increment routing metrics. Solutions include Split Horizon and Poison Reverse.

Unit 4: Transport Layer Protocols & Congestion Control

Key Syllabus Topics: User Datagram Protocol (UDP), Transmission Control Protocol (TCP), 3-Way Handshake Connection Establishment, TCP Teardown, TCP Sliding Window Flow Control, Congestion Control (Slow Start, Congestion Avoidance, Fast Retransmit, Fast Recovery).

Theoretical Framework & Concepts:

Transport layer provides logical process-to-process communication using Port numbers. TCP guarantees reliable, in-order byte stream delivery with dynamic congestion window management (AIMD).

Implementation / Key Formula Reference:

```
// TCP 3-Way Handshake: SYN -> SYN-ACK -> ACK
```

Frequently Asked University Exam Questions:

• **Q1: Differentiate between TCP and UDP with header structures and use cases.**

Solution: A: TCP is connection-oriented, reliable, with flow/congestion control (20-byte header; used for HTTP, SSH). UDP is connectionless, lightweight, and low-latency (8-byte header; used for DNS, VoIP, streaming).

Unit 5: Application Layer & Network Security

Key Syllabus Topics: Domain Name System (DNS), HTTP/1.1 vs HTTP/2 vs HTTP/3, HTTPS & TLS Handshake, Email Protocols (SMTP, POP3, IMAP), Symmetric/Asymmetric Encryption, Digital Signatures, Firewalls & VPNs.

Theoretical Framework & Concepts:

Application layer protocols define syntax and message exchanges between client and server applications. HTTPS encrypts HTTP traffic via TLS symmetric session keys exchanged during asymmetric handshake.

Implementation / Key Formula Reference:

```
GET /index.html HTTP/1.1
Host: pbrvits.ac.in
User-Agent: StudentCompanion/1.0
```

Frequently Asked University Exam Questions:

- **Q1: Explain the recursive and iterative DNS resolution workflow.**

Solution: A: Client queries Local DNS. In recursive resolution, the server resolves all referrals; in iterative resolution, root/TLD servers return next authoritative name server referrals to the resolver.